This is a one-page summary introducing the central motivation and outstanding issues of the Smart Radon Fan (SmRF).

Figure 1 depicts the dilemma of always-on, fixed-speed radon fan mitigation. There may be lengthy periods when mitigation is not required or can be accomplished with significantly less than a 100% fan utilization.

Figure 2 depicts the dilemma of short-term (72-hour) testing. There may be false negatives which mislead the homeowner, as well as true positives that indicate the need for mitigation, but in an environment where the properties depicted in Figure 1 occur.

These two well-known cases lead to a continuously monitored, variable-speed radon fan mitigation solution. In this vein, statistical models indicate frequent CRM readings, even if somewhat noisy, are preferred to intermittent and short-term accurate readings, given the general variability in actual radon levels.

Figure 3 depicts the benefits of a continuously monitored, variable-speed radon fan mitigation solution. An average radon level target (here 2.0 pCi/L) is achieved with fan utilization at approximately 50%; and there are no extreme deviations from the average value.

However, there is a problem with the residential, continuously monitored solution. If there is significant drift in consumer-grade, uncalibrated CRMs (e.g., a negative bias), then variable-speed fan mitigation may be inadequate, and negative radon level readings will give a false sense of successful mitigation.

The drift problem can be addressed in two ways. (1) When using a fixed-position CRM with a variable-speed fan, it might be possible to detect drift by detecting long-term differences in fan utilization, and this is under investigation. (2) Should consumer-grade CRM manufacturers publish data on the drift properties of their devices, replacement schedules of relatively inexpensive CRMs can address the drift problem.

As examples of the last point, Airthings believes their monitors have a built-in drift correction using a variety of radon decay counting methods. Aranet has published drift models that give radon level margin of error as a function of total exposure, when radon progeny elements may continue to accumulate (and generate alpha emissions) in the CRM.

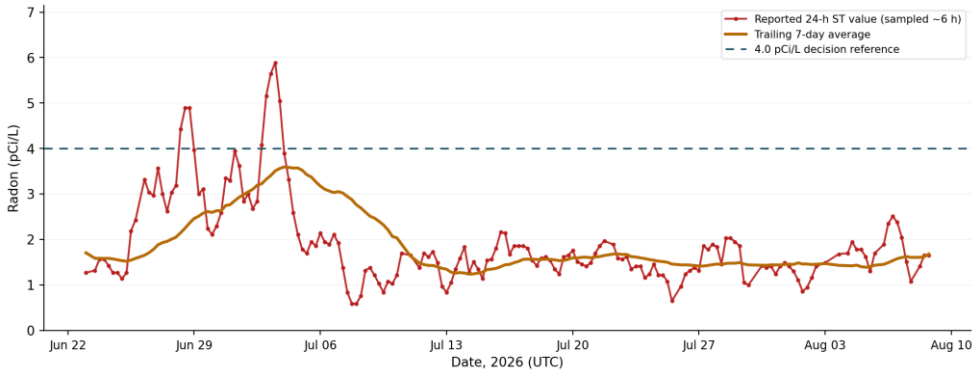


Figure 1

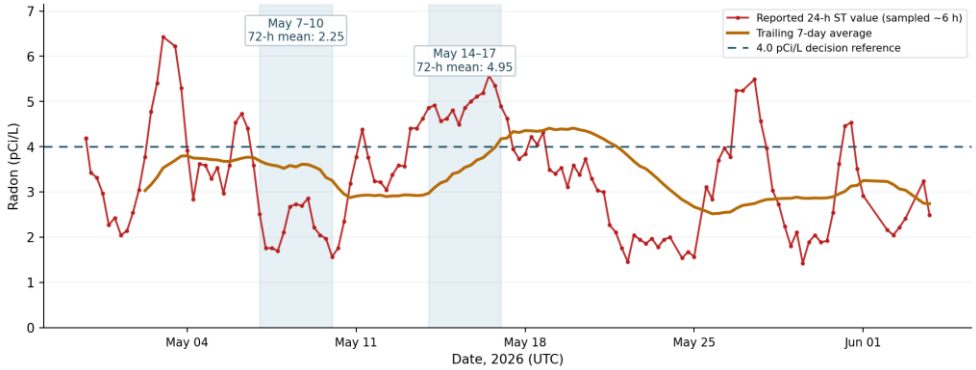


Figure 2

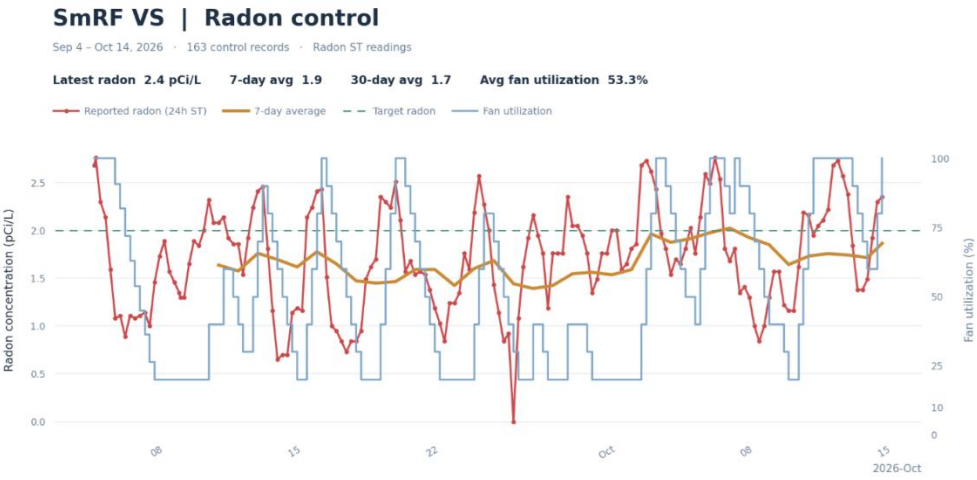


Figure 3